

torch.nn.functional.fractional_max_pool3d

https://pytorch.org/docs/stable/generated/torch.nn.functional.fractional_max_pool3d.html

Description:

Applies 3D fractional max pooling over an input signal composed of several input planes. Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham. The max-pooling operation is applied in $k_T \times k_H \times k_W$ regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes. `kernel_size` – the size of the window to take a max over. Can be a single number k (for a square kernel of $k \times k \times k$) or a tuple (k_T, k_H, k_W) . `output_size` – the target output size of the form $o_T \times o_H \times o_W$. Can be a tuple (o_T, o_H, o_W) or a single number o for a cubic output.

Function Signature

```
torch.nn.functional.fractional_max_pool3d(input,  
kernel_size, output_size=None, output_ratio=None,  
return_indices=False, _random_samples=None)[source]#
```

Parameters

Shape:

Examples::

Code Examples

```
>>> input = torch.randn(20, 16, 50, 32, 16)  
>>> # pool of cubic window of size=3, and target output size  
13x12x11  
>>> F.fractional_max_pool3d(input, 3, output_size=(13, 12, 11))  
>>> # pool of cubic window and target output size being half of  
input size  
>>> F.fractional_max_pool3d(input, 3, output_ratio=(0.5, 0.5,  
0.5))
```

Detailed Documentation

Documentation

Applies 3D fractional max pooling over an input signal composed of several input planes.

Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham

The max-pooling operation is applied in $k_T \times k_H \times k_W$ regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes.

`kernel_size` – the size of the window to take a max over. Can be a single number k (for a square kernel of $k \times k \times k$) or a tuple (k_T, k_H, k_W)

`output_size` – the target output size of the form $o_T \times o_H \times o_W$. Can be a tuple (o_T, o_H, o_W) or a single number o for a cubic output $o \times o \times o$

`output_ratio` – If one wants to have an output size as a ratio of the input size, this option can be given. This has to be a number or tuple in the range $(0, 1)$

`return_indices` – if True, will return the indices along with the outputs. Useful to pass to `max_unpool3d()`.

Input: $(N, C, T_{in}, H_{in}, W_{in})$ or $(C, T_{in}, H_{in}, W_{in})$

Output: $(N, C, T_{out}, H_{out}, W_{out})$ or $(C, T_{out}, H_{out}, W_{out})$, where $(T_{out}, H_{out}, W_{out}) = \text{output_size}(T_{out}, H_{out}, W_{out}) = \text{output_size}(T_{out}, H_{out}, W_{out})$ or $(T_{out}, H_{out}, W_{out}) = \text{output_ratio} \times (T_{in}, H_{in}, W_{in})$

